TASK – 2

**Design and Implementation of a Real-Time Stream Analysing Engine Using Stream Processing Concepts Java**

PROGRAM :

```
import java.util.*;
import java.util.stream.*;

public class StreamAnalysisEngine {
   public static void main(String[] args) {

      List<Integer> dataStream = Arrays.asList(
         10, 25, 40, 15, 60, 80, 30, 100
      );

      System.out.println("Incoming Data Stream:");
      dataStream.forEach(System.out::println);

      List<Integer> filteredData = dataStream.stream()
         .filter(value -> value >= 30)
         .collect(Collectors.toList());

      long count = filteredData.stream().count();

      int sum = filteredData.stream()
         .mapToInt(Integer::intValue)
         .sum();
```

```
        double average = filteredData.stream()
            .mapToInt(Integer::intValue)
            .average()
            .orElse(0.0);

        System.out.println("\n--- Stream Analysis Results ---");
        System.out.println("Filtered Data: " + filteredData);
        System.out.println("Count: " + count);
        System.out.println("Sum: " + sum);
        System.out.println("Average: " + average);

    }
}
```

INPUT :

10, 25, 40, 15, 60, 80, 30, 100

OUTPUT :

```
Incoming Data Stream:
10
25
40
15
60
80
30
100

--- Stream Analysis Results ---
```

Filtered Data: [40, 60, 80, 30, 100]

Count: 5

Sum: 310

Average: 62.0